MEGATHON X: SINGULARITY · PROBLEM STATEMENT 2

Green Home Score and Retrofit Recommendation Engine



How green is my home today, what should I improve, how much could I save, and could it add value?

Lloyds

Hackathon briefing for participants

01 · THE CHALLENGE

What you are building

Build an **AI system** that analyses property characteristics, energy consumption and existing home features to:

1. Calculate a **Green Home Score**
2. Identify homes with **poor energy efficiency**
3. **Recommend suitable improvements**: insulation, heat pumps, solar panels, double glazing, smart energy controls
4. Estimate potential **energy savings and carbon reduction**
5. Estimate the potential **increase in property value**

How green is my home today?

What should I improve?

How much energy and carbon could I save?

Could this increase my property's value?

02 · THE CORE RULE

Models calculate. The LLM explains.

Models and calculations

The **score, savings, carbon and value** figures all come from documented models and calculations.

The LLM

Only **explains** outputs and answers what-if questions. It must not calculate the score or invent estimates.

What-if requests

First **rerun** the relevant models and calculations for the proposed scenario. Then the LLM explains their outputs.

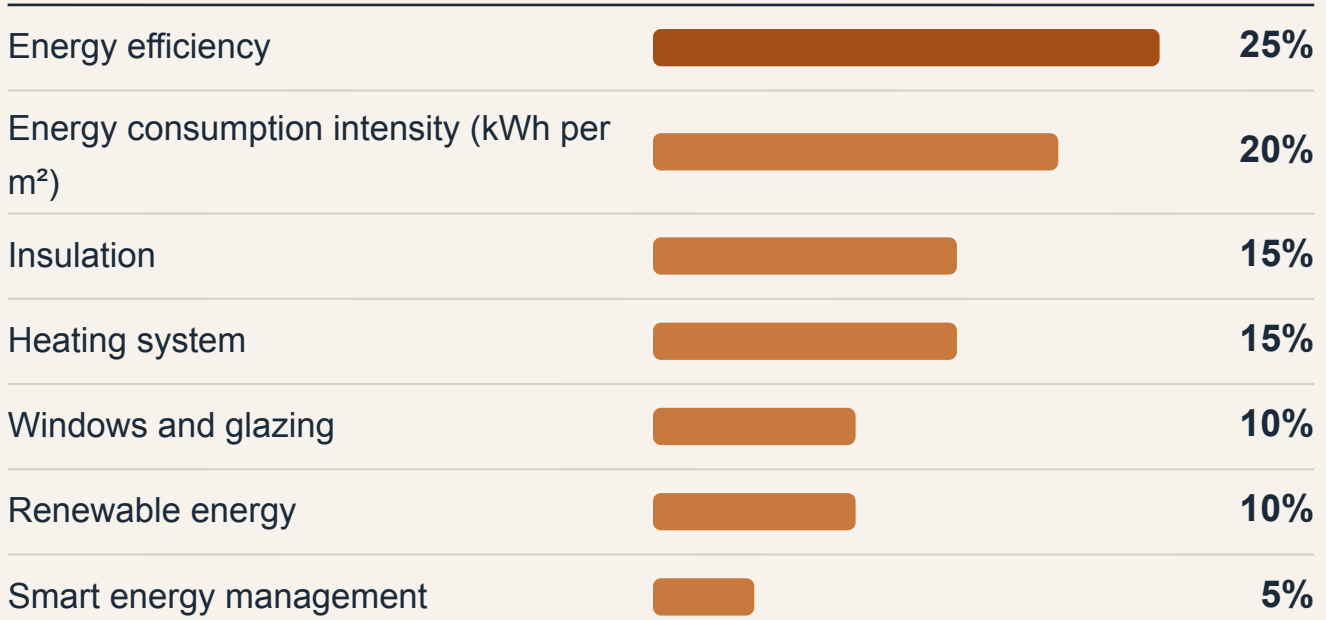
A 0–100 score from seven weighted components

$$S = \Sigma\ w_k \times s_k$$

s_k: component sub-score, 0 to 100

w_k: fixed weight as a fraction, Σ w_k = 1

Score	Classification
80–100	Highly efficient
65–79	Good
50–64	Moderate
30–49	Poor
0–29	Very poor



Energy efficiency	25%
Energy consumption intensity (kWh per m²)	20%
Insulation	15%
Heating system	15%
Windows and glazing	10%
Renewable energy	10%
Smart energy management	5%
Total	**100%**

04 · RANKING BY IMPROVEMENT POTENTIAL

How much purchased energy could feasible retrofits remove?

$$\text{Improvement potential} = 100 \times (E_{baseline} - E_{after}) / E_{baseline}$$

What it measures

Estimated % reduction in annual purchased energy achievable with the retrofits feasible for that property.

Purchased energy

Grid electricity imported + gas consumed, in kWh/year. Both totals use the same accounting convention.

Missing baseline

If baseline energy is zero or unavailable, report the metric as unavailable and document the handling.

Per fuel

Report electricity and gas changes separately: a retrofit may cut one and raise the other. Solar reduces grid imports, not underlying demand.

05 · RETROFIT RECOMMENDATIONS

A prioritized list of what is feasible

1

Feasible, not exhaustive

Return the retrofits that are feasible for this property, not every retrofit available.

2

Every item explains itself

Each item gives its suitability, estimated saving and reason.

High Medium Low

3

Order reflects interactions

Sequence retrofits by how they interact, for example:

Insulation → Heat pump

Per retrofit, per package, per fuel

- Estimate the saving for **each recommended retrofit** and for **the package** as a whole.
- In **kWh/year**, separately for **electricity and gas**.
- Use a **consistent baseline** and estimate post-retrofit consumption for each fuel.
- Show an increase in consumption as a **negative saving**.
- For a package, **account for interactions** rather than simply adding individual estimates.
- Label all savings as **estimates, not guarantees**.

ILLUSTRATIVE AGGREGATE EXAMPLE

Baseline electricity	4,800 kWh/yr
+ Baseline gas	12,000 kWh/yr
= Purchased energy	16,800 kWh/yr
× 15% reduction	**2,520 kWh/yr**

This aggregate does not determine the electricity and gas savings: your model must estimate each fuel separately. A full package might, for example, net 5,500 kWh/year after interactions.

Carbon is calculated fuel by fuel

$$\Delta C = \Sigma\ \Delta E_f \times EF_f,\ f \in \{elec, gas\}$$

$$= \Delta E_{elec} \times EF_{elec} + \Delta E_{gas} \times EF_{gas}$$

ΔE_f

Change in consumption of fuel f: E_f,baseline − E_f,after, in kWh/year.

EF_f

Emissions factor for fuel f, in kg CO_2e per kWh, from a reliable, documented source with units.

Signs

Positive ΔE_f is a saving; negative is an increase. Positive ΔC is an emissions reduction.

Why per fuel? A heat pump may lower gas use while raising electricity use.

Give a range, never a single number

Current value
£350,000

→

Energy rating
E to C

→

Indicative uplift
2–5%

→

Potential value
£357,000 to £367,500

- Describe it as an **estimated potential** property-value impact, based on historical and market relationships in your dataset.
- Compare a **consistent baseline** with the post-retrofit scenario, e.g. predicted values for the same property before and after, other characteristics held constant.
- Account for **location, size, property type and market conditions**.
- If your data and method can’t establish a causal retrofit effect, call it an **indicative estimate**, not a guaranteed or proven causal increase.

09 · PIPELINE AND WHAT-IF FLOW

From raw data to the customer dashboard

1. Property, energy and sales data
2. Feature engineering
3. Green Home Score
4. Efficiency detector
5. Retrofit engine
6. Savings, carbon and value models
7. LLM explanation
8. Customer dashboard

WHAT-IF FLOW · THE LLM MUST NOT INVENT SCENARIO OUTPUTS

Request updates the proposed retrofit scenario → Rerun the savings, carbon and value calculations → LLM explains the computed results

10 · DATASET

You prepare the data your solution needs

Where it can come from

- Synthetic property, energy, retrofit and sales data
- Suitable public datasets that may lawfully be used
- A combination of both

No real personal or confidential customer data. Use public data in line with its licence and restrictions.

What you must document

- Data sources and schema
- Assumptions and limitations
- Synthetic-data generation and label rules; emissions factors with units and sources

Room to adapt

- Construct plausible reference labels for improvement-potential ranking, retrofit feasibility, per-fuel savings and property-value impact. Your data must support every required output.

11 · OPTIONAL EXTENSION

Customer memory

Give each customer a **persistent memory** and use it to adapt later recommendations. The design is open. It could hold:

- **Completed retrofits**
- **Stated constraints or preferences**
- **Past recommendations and the customer’s responses**

GUARDRAIL

Scores and estimates must still come from the models and calculations.

IF YOU ATTEMPT IT

Show how recommendations change as memory builds up.

12 · EVALUATION

How your solution will be tested

Your methodology

- Documented train, validation and test splits, where applicable
- Report methodology, metrics, baseline comparisons and limitations
- Data quality, label generation, validation and reproducibility count alongside results

Scenario tests to expect

- Missing or conflicting property data
- Infeasible or interacting retrofits
- Higher electricity use after a heating-system change
- Too little information to estimate savings or value impact

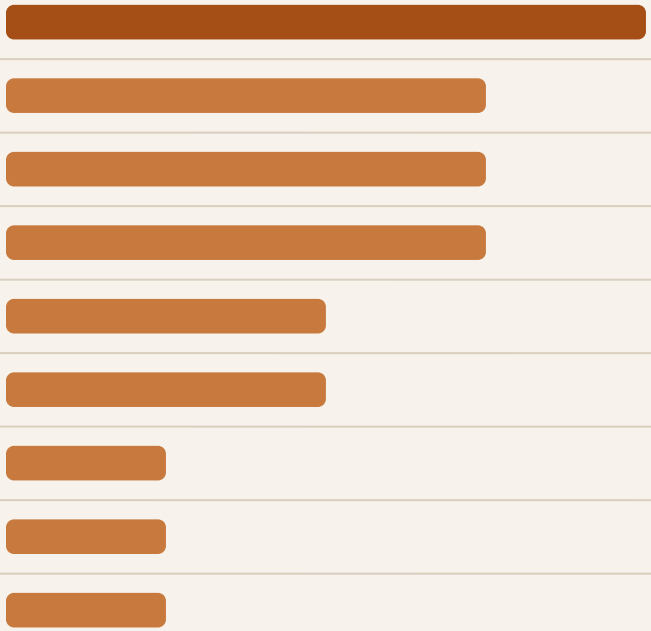
Baseline and reproducibility

- Compare with an appropriate simple baseline on your own validation data, and explain the results
- Organisers may rerun your code and inspect data, labels, calculations and outputs
- Results that can’t be reproduced may affect your evaluation

⚠ Raw metrics are not a standalone ranking criterion. LLM targets: ≥90% groundedness, ≥90% factual accuracy, and <5% unsupported claims.

Score Distribution

ML and recommendation quality (data quality, validation methodology and held-out metrics)	20%
Green Home Score and sustainability insight quality	15%
Energy savings, carbon reduction and property-value estimation	15%
GenAI and LLM implementation (grounded explanations and what-if evaluation)	15%
Business relevance and usefulness of the recommendations	10%
Technical architecture (including data engineering, reproducibility and system performance)	10%
Explainability and Responsible AI	5%
UX and dashboard	5%
Innovation (including the optional extension)	5%



Unsupported precision, arbitrary synthetic labels or overclaiming may reduce your score.

Nine things to hand in

A	Working prototype (web app or dashboard)
B	Source code and data preparation/generation scripts, with run instructions; dataset files where sharing is permitted
C	Trained models and inference code
D	Scoring method: documented sub-score mappings
E	LLM implementation: prompts, model or API details, evaluation approach
F	Architecture diagram
G	Evaluation report: the evaluation metrics, on the validation set
H	Responsible AI note: privacy, bias, explainability, hallucination, human oversight, data limitations
I	Held-out test predictions, with the data schema and label documentation